

# THE INTERNET OF THINGS



News, views and development on the Internet of Things

## “Gesture interfaces would act as a catalyst for other modes of interaction...”

Satish Patil is chief designer—Use Experience, Tata Elxsi. He speaks to Janani Gopalakrishnan Vikram about the importance of the user interface in making communication with devices more natural, in an IoT era that promises a tech-filled environment, which can sometimes be intimidating

**The ‘Internet of Things’ conjures up images of an environment where all objects are connected to the Internet, and are visibly or invisibly collecting information, watching what we do and helping us. So many connected devices might also end up intimidating people! How does gesture-enabled technology make IoT devices more user friendly?**

The last 50-60 years have seen rapid changes in the way we interact with devices and systems. There has been an evolution in input technologies from keyboards and touchscreen devices to voice, gestures and, literally, mind-controlled devices. Technology is in a constant state of flux creating new use-case scenarios and experiences, which continue to more closely map human actions and thought processes, making devices ‘more natural’.

From text entry strings and graphical user interfaces, on to touchscreen interfaces, we have come a long way. Currently, voice and gesture interfaces are being explored to make interactions with devices almost similar to interacting with humans, like asking your phone to dial your friend, for instance.

While generic consumer electronics will find its usages and applications, gestures as a mode of interaction would be used far more effectively in certain special circumstances. For instance, in automotive interfaces, where there is a need to keep looking at the road

even as you use the technology in the car. Gestures could possibly reduce the cognitive load on the driver. Similarly, gesture interfaces will increasingly play a vital role in making products that differently abled people can use.

**Can you tell us about the gesture-based automotive human-machine interface (HMI) recently developed by you for a client?**

Typical automotive HMIs are always ‘multi-modal’, employing multiple modes of interaction such as touch, voice or gestures, in tandem with physical controls.

We have helped automotive clients to define gesture interfaces by devising a meaningful repository of gestures to accomplish certain functions in cars. For instance, we have developed a gesture-based HMI that allows a driver to browse through a playlist or address book and play a song or make a call.

Additionally, to do a visual design for the application, based on gesture interface, has its own challenges. It is important that while designing the screens and controls, their layout and visual design elements should be ‘in sync’ with the gesture and its output. Also, in dual interaction modes, wherein functionalities can be handled by both touch and gestures, it is challenging to do the visual design of interfaces. One has to make sure the users are aware of the interaction modes, and the design it-

self supports both the touch and gesture behaviour separately, in its own ways.

**The main need for gesture-based input is to make interaction with a device more natural. However, I guess it would not be so natural if the output was always coming as text on a screen? So, what do you think are the possible means to make the response from the device equally natural?**

Yes, the output of gesture interaction need not be limited to a change of status or confirmation of action on the screen. But as I mentioned earlier, the gestures deployed will lead to direct ‘actions’ with or without the notification on screens. For instance, in mobile phones, simply shaking the phone can help you change or play the next song. In cars, a simple hand gesture near the AC panel can change the temperature settings. Some of the console-based applications can virtually act as a ‘gym’ for you, wherein the applications based on gesture interactions enable you to do extensive physical activities through multiple stimuli, like sports activities and games.

Having said this, it is also important to note that like some of the other interface techniques, gesture-based interfaces are evolving to provide a complete user experience. They could act as a catalyst for other modes of interaction, rather than being the sole mode of interaction with the product.